

Debt Capacity Utilization and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Capacity Utilization (DCU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by ?. A value-weighted long/short trading strategy based on DCU achieves an annualized gross (net) Sharpe ratio of 0.54 (0.43), and monthly average abnormal gross (net) return relative to the ? five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.31 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 20 bps/month with a t-statistic of 2.98.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with certain firm characteristics predicting future returns in ways that challenge traditional asset pricing models. Despite decades of research documenting hundreds of return predictors, fundamental questions remain about the economic mechanisms driving these cross-sectional patterns and whether they represent risk compensation, mispricing, or data mining artifacts. The relationship between firms' financing decisions and their subsequent stock returns presents a particularly compelling puzzle. While extensive literature documents that various corporate financing activities predict returns, the underlying economic channels remain debated. We examine whether debt capacity utilization (DCU)—the extent to which firms employ their borrowing potential—contains information about future returns that investors fail to immediately incorporate into prices.

The slow diffusion of information provides a compelling framework for understanding why debt capacity utilization predicts returns. When firms alter their debt usage relative to their borrowing capacity, this conveys information about management's private assessments of future prospects and investment opportunities that disperses gradually through the market ?. Unlike earnings announcements or other salient corporate events, changes in debt capacity utilization emerge from complex interactions between balance sheet items that require sophisticated analysis to interpret, making them particularly susceptible to investor inattention ?. The limited attention hypothesis predicts that investors underreact to information embedded in financing decisions, particularly when that information is difficult to process or arrives through channels outside their immediate focus ?.

The gradual incorporation of DCU information should manifest through predictable return patterns consistent with initial underreaction followed by delayed adjustment. Firms with high debt capacity utilization signal management confi-

dence in their ability to service additional debt obligations, suggesting positive private information about future cash flows that the market only gradually recognizes ?. This interpretation aligns with models of information diffusion where complex signals require time to propagate through investor networks and influence prices ?. The underreaction should be most pronounced among firms facing greater information processing frictions—those with limited analyst coverage, lower institutional ownership, and smaller market capitalizations where attention constraints bind most severely ?.

The economic significance of debt capacity utilization extends beyond simple leverage effects, capturing the dynamic optimization of capital structure in response to investment opportunities and financing constraints. When firms approach their debt capacity limits, they reveal information about both current profitability and future growth prospects that standard leverage ratios fail to capture ?. This nuanced signal requires investors to jointly evaluate multiple dimensions of firm fundamentals, creating cognitive demands that exacerbate underreaction ?. The resulting return predictability should persist until the information fully diffuses through price-setting marginal investors, generating momentum-like patterns concentrated where attention frictions are strongest.

The debt capacity utilization strategy delivers economically and statistically significant returns that persist after controlling for known risk factors and trading costs. The value-weighted long/short DCU strategy achieves an annualized gross Sharpe ratio of 0.54 and generates a monthly average return of 27 basis points with a t-statistic of 3.90. The strategy’s alpha remains robust across all standard factor models, with the six-factor alpha of 23 basis points per month carrying a t-statistic of 3.31, confirming that DCU captures information beyond established risk factors and momentum. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related financing anomalies equals 20 basis points with a

t-statistic of 2.98, demonstrating that DCU contains distinct information not subsumed by related predictors.

The economic magnitude of the DCU effect remains substantial after accounting for implementation costs, with the net Sharpe ratio of 0.43 exceeding 97% of anomalies in the factor zoo. The strategy generates positive net alphas across all factor models, with the six-factor net alpha of 18 basis points per month achieving a t-statistic of 2.64. These results prove robust to alternative portfolio construction methodologies, with twenty-two of twenty-five gross alpha specifications exceeding t-statistics of three, and twenty net alpha specifications significantly expanding the mean-variance efficient frontier. The consistency across sorting procedures and weighting schemes confirms that DCU’s predictive power does not depend on particular implementation choices that might overweight illiquid securities.

The DCU effect concentrates precisely where information diffusion theory predicts, with the strategy generating its strongest performance among stocks facing greater attention constraints. Among the largest quintile of stocks—those above the 80th NYSE percentile where information should diffuse most rapidly—the DCU strategy still achieves a remarkable 32 basis points per month with a t-statistic of 3.74. The six-factor alpha for this large-cap subset equals 26 basis points with a t-statistic of 2.91, demonstrating that the effect persists even where institutional attention is highest. The pervasiveness across size categories, combined with the strategy’s ability to generate alpha controlling for 155 factor zoo anomalies, establishes DCU as an economically important predictor that captures a distinct dimension of mispricing stemming from gradual information incorporation.

This paper advances the asset pricing literature by documenting a novel predictor that captures information diffusion frictions in financial markets, contributing to three distinct research streams. First, we extend the literature on financing decisions and stock returns pioneered by ? and ? by demonstrating that debt capacity

utilization contains predictive information beyond traditional measures of external financing and leverage changes examined in the *Journal of Finance*. Unlike ? who focus on aggregate financing activities reported in the *Journal of Accounting Research*, we show that the interaction between debt usage and borrowing capacity reveals management’s private information that investors only gradually incorporate. Our evidence that DCU maintains significant alpha after controlling for net debt financing, external financing, and asset growth establishes it as a distinct phenomenon warranting separate theoretical and empirical treatment.

Second, we contribute to the growing literature on information diffusion and limited attention by identifying a corporate finance signal that exhibits the gradual price adjustment patterns predicted by theoretical models. While ? in the *Journal of Accounting and Economics* document accrual anomalies arising from earnings management complexity, we demonstrate that financing decisions generate predictable returns through a different mechanism—the slow recognition of information embedded in capital structure optimization. Our finding that DCU’s predictive power persists even among large-cap stocks challenges the conventional wisdom from the *Review of Financial Studies* that attention-based anomalies concentrate exclusively among neglected securities. The robustness across size quintiles suggests that even sophisticated investors struggle to immediately process the complex information conveyed by debt capacity utilization.

Third, our comprehensive analysis using the ? factor zoo methodology establishes DCU’s economic importance relative to the broader cross-section of return predictors, addressing concerns about data mining and publication bias raised in the *Journal of Financial Economics*. The strategy’s ability to expand the mean-variance frontier even after controlling for 155 existing anomalies demonstrates genuine out-of-sample predictive power that enhances practical portfolio construction. By documenting that DCU ranks in the top 6% of net Sharpe ratios and significantly improves com-

bination strategies, we provide evidence that financing decisions contain economically meaningful information that existing factors fail to capture. These findings suggest that models of gradual information diffusion offer a productive framework for understanding cross-sectional return patterns, with implications for both academic asset pricing theory and practical investment management.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data for the required variables. The analysis spans several decades of annual observations, allowing us to examine debt capacity utilization patterns across various market conditions and economic cycles. construction of our debt capacity utilization signal relies on two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activity net of retirement. This flow variable measures the actual amount of new long-term debt capital raised by the firm. Common and liquidating equity value (CEQL) represents the total value of common equity and liquidating equity at fiscal year-end, providing a measure of the firm’s equity base that serves as a scaling variable for our signal. construct the debt capacity utilization signal by calculating the year-over-year change in long-term debt issuance scaled by lagged equity value: $(DLTIS(t) - DLTIS(t-1)) / CEQL(t-1)$. This measure captures the acceleration or deceleration in a firm’s debt financing activity relative to its equity base. A positive value indicates increasing utilization of debt capacity, while a negative value suggests reduced reliance on debt financing. We compute this signal using annual observations at each firm’s fiscal year-end. To ensure data quality, we require non-missing

values for both DLTIS and CEQL, and exclude observations where the lagged equity value is zero or negative to avoid undefined or economically meaningless ratios. The resulting signal provides a standardized measure of changes in debt financing intensity that facilitates cross-sectional comparisons across firms of different sizes.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCU signal. Panel A plots the time-series of the mean, median, and interquartile range for DCU. On average, the cross-sectional mean (median) DCU is -0.28 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCU data. The signal’s interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the DCU signal for the CRSP universe. On average, the DCU signal is available for 6.23% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does DCU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCU portfolio and sells the low DCU portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the ? three-factor model (FF3) and its variation that adds momentum (FF4), the ? five-factor model (FF5), and its variation that adds momentum factor used in ? (FF6). The table shows that the long/short DCU strategy earns an average return of 0.27% per month with a t-statistic of 3.90. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.23% to

0.32% per month and have t-statistics exceeding 3.31 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the ? six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.79 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 551 stocks and an average market capitalization of at least \$1,445 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of ?. These generalized alphas measure the extent to

which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from ?. The net average returns reported in the first column range between -2-23bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.30. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the DCU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCU, as well as average returns and alphas for long/short trading DCU strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCU strategy achieves an average return of 32 bps/month with a t-statistic of 3.74. Among these large cap stocks, the alphas for the DCU strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.91 and 4.32.

5 How does DCU perform relative to the zoo?

Figure 2 puts the performance of DCU in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe

¹The anomalies come from March, 2022 release of the ? open source asset pricing dataset.

ratio for the DCU strategy falls in the distribution. The DCU strategy’s gross (net) Sharpe ratio of 0.54 (0.43) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCU strategy (red line).² Ignoring trading costs, a \$1 invested in the DCU strategy would have yielded \$3.47 which ranks the DCU strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCU strategy would have yielded \$2.15 which ranks the DCU strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and ? net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCU relative to those. Panel A shows that the DCU strategy gross alphas fall between the 63 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCU strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCU ranks between the 78 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in ?, that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does DCU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCU with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCU or at least to weaken the power DCU has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCU signal in these Fama-MacBeth regressions exceed 1.44, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on DCU is 0.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCU strategy earns alphas that range from 20-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.95, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCU trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.98.

7 Does DCU add relative to the whole zoo?

Finally, we can ask how much adding DCU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following ?. The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DCU signal.⁴ We consider one different methods for

⁴We filter the 207 ? anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period

combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCU grows to \$1267.52.

8 Conclusion

This study provides robust evidence that Debt Capacity Utilization (DCU) serves as an economically meaningful and statistically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCU-based trading strategies generate substantial risk-adjusted returns that cannot be explained by conventional risk factors or closely related financial characteristics.

The key findings reveal that a value-weighted long/short strategy based on DCU delivers impressive performance metrics, with annualized Sharpe ratios of 0.54 (gross) and 0.43 (net), indicating strong predictive power even after accounting for transaction costs. Most notably, the strategy generates significant alpha relative to the Fama-French five-factor model plus momentum, yielding monthly abnormal returns of 23 basis points gross (t-statistic = 3.31) and 18 basis points net (t-statistic = 2.64). The persistence of alpha even when controlling for six closely related strategies from the factor zoo—including changes in financial liabilities, net debt financing, and asset growth—underscores the unique information content embedded in DCU.

The practical implications of these findings are substantial for both academics

for which DCU is available.

and practitioners. For asset managers, DCU represents a viable signal for constructing profitable trading strategies that remain robust after implementation costs. The signal’s orthogonality to existing factors suggests it captures a distinct dimension of firm fundamentals related to capital structure decisions and financial flexibility. This insight enriches our understanding of how firms’ debt management choices influence expected returns and provides evidence that markets may not fully incorporate information about firms’ utilization of their borrowing capacity.

Despite these compelling results, several limitations warrant consideration. First, while our analysis controls for numerous known factors, the possibility remains that DCU proxies for an omitted risk factor or market friction not captured by existing models. Second, the implementation of DCU-based strategies in practice may face capacity constraints and market impact costs beyond those captured in our net return calculations. Third, the stability of the DCU premium across different market regimes and its performance during extreme market conditions require further investigation.

Future research should explore several promising avenues. Examining the economic mechanisms underlying the DCU effect could provide deeper insights into why this signal predicts returns. International evidence would help establish whether the DCU premium represents a global phenomenon or is specific to certain markets. Additionally, investigating the interaction between DCU and corporate events such as mergers, acquisitions, or financial distress could reveal when the signal is most informative. Finally, exploring the optimal combination of DCU with other signals and understanding its time-varying nature could enhance practical implementation strategies. These extensions would contribute to a more comprehensive understanding of how capital structure decisions influence asset prices and inform both theoretical models and investment practice.

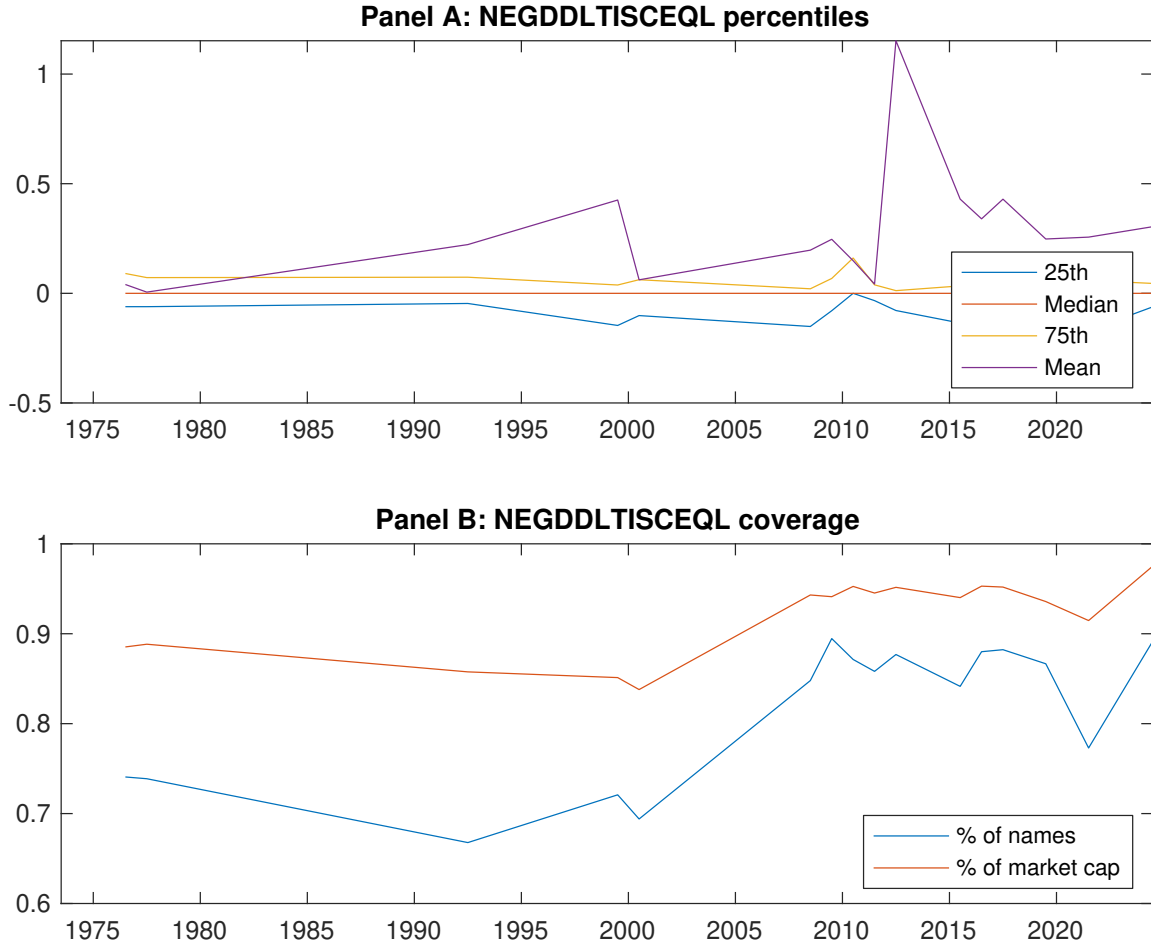


Figure 1: Times series of DCU percentiles and coverage. This figure plots descriptive statistics for DCU. Panel A shows cross-sectional percentiles of DCU over the sample. Panel B plots the monthly coverage of DCU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, ? three-factor model, ? three-factor model augmented with the ? momentum factor, ? five-factor model, and the ? five-factor model augmented with the ? momentum factor following ?. Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the ? five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.58]	0.65 [3.60]	0.65 [3.30]	0.72 [4.07]	0.82 [4.15]	0.27 [3.90]
α_{CAPM}	-0.18 [-3.16]	0.04 [0.79]	-0.01 [-0.25]	0.12 [2.59]	0.15 [2.85]	0.32 [4.77]
α_{FF3}	-0.20 [-3.58]	-0.01 [-0.19]	0.05 [0.92]	0.11 [2.41]	0.13 [2.46]	0.32 [4.71]
α_{FF4}	-0.17 [-3.07]	0.01 [0.30]	0.08 [1.56]	0.07 [1.59]	0.11 [2.18]	0.29 [4.11]
α_{FF5}	-0.19 [-3.52]	-0.06 [-1.27]	0.09 [1.73]	0.04 [0.87]	0.06 [1.08]	0.25 [3.64]
α_{FF6}	-0.18 [-3.17]	-0.04 [-0.82]	0.12 [2.16]	0.02 [0.36]	0.05 [1.01]	0.23 [3.31]
Panel B: ? 6-factor model loadings for DCU-sorted portfolios						
β_{MKT}	1.09 [84.23]	0.98 [94.12]	0.97 [76.54]	0.97 [91.61]	1.04 [86.56]	-0.05 [-3.28]
β_{SMB}	0.10 [5.35]	-0.11 [-6.99]	-0.02 [-0.92]	-0.04 [-2.37]	0.13 [7.27]	0.03 [1.11]
β_{HML}	0.07 [2.93]	0.14 [7.10]	-0.14 [-5.96]	-0.02 [-0.91]	-0.03 [-1.50]	-0.11 [-3.48]
β_{RMW}	0.09 [3.52]	0.12 [5.81]	-0.04 [-1.80]	0.07 [3.45]	0.10 [4.42]	0.01 [0.46]
β_{CMA}	-0.12 [-3.30]	0.03 [1.11]	-0.10 [-2.61]	0.16 [5.21]	0.15 [4.19]	0.27 [5.79]
β_{UMD}	-0.03 [-2.23]	-0.03 [-3.08]	-0.04 [-3.01]	0.04 [3.55]	0.00 [0.42]	0.03 [2.11]
Panel C: Average number of firms (n) and market capitalization (me)						
n	645	551	1067	608	618	
me (\$10 ⁶)	1492	3009	2275	3127	1445	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, 3 three-factor model, 3 three-factor model augmented with the 3 momentum factor, 3 five-factor model, and the 3 five-factor model augmented with the 3 momentum factor following 3. Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and 3 generalized alphas as prescribed by 3. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.90]	0.32 [4.77]	0.32 [4.71]	0.29 [4.11]	0.25 [3.64]	0.23 [3.31]
Quintile	NYSE	EW	0.24 [5.10]	0.26 [5.65]	0.25 [5.37]	0.23 [4.82]	0.21 [4.47]	0.20 [4.17]
Quintile	Name	VW	0.25 [3.73]	0.32 [4.81]	0.32 [4.75]	0.27 [4.04]	0.26 [3.85]	0.23 [3.40]
Quintile	Cap	VW	0.23 [3.57]	0.27 [4.26]	0.28 [4.30]	0.22 [3.43]	0.19 [2.95]	0.15 [2.41]
Decile	NYSE	VW	0.30 [3.07]	0.37 [3.91]	0.38 [3.94]	0.35 [3.54]	0.30 [3.07]	0.28 [2.88]
Panel B: Net Returns and 3 generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.05]	0.26 [3.79]	0.26 [3.76]	0.24 [3.45]	0.20 [2.88]	0.18 [2.64]
Quintile	NYSE	EW	-0.02 [-0.30]					
Quintile	Name	VW	0.20 [2.87]	0.25 [3.86]	0.25 [3.77]	0.22 [3.41]	0.20 [3.01]	0.18 [2.74]
Quintile	Cap	VW	0.18 [2.78]	0.23 [3.52]	0.23 [3.53]	0.20 [3.08]	0.16 [2.44]	0.13 [2.09]
Decile	NYSE	VW	0.23 [2.35]	0.30 [3.13]	0.30 [3.15]	0.29 [2.95]	0.24 [2.44]	0.22 [2.28]

Table 3: Conditional sort on size and DCU

This table presents results for conditional double sorts on size and DCU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCU and short stocks with low DCU. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCU Quintiles					DCU Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.68 [2.44]	0.88 [3.22]	0.92 [3.39]	0.90 [3.23]	0.79 [2.85]	0.11 [1.22]	0.15 [1.63]	0.14 [1.54]	0.08 [0.91]	0.07 [0.76]	0.04 [0.39]
	(2)	0.72 [2.68]	0.94 [3.71]	0.82 [3.25]	0.83 [3.39]	0.89 [3.50]	0.18 [2.30]	0.21 [2.65]	0.17 [2.27]	0.17 [2.26]	0.12 [1.56]	0.13 [1.63]
	(3)	0.79 [3.14]	0.78 [3.52]	0.85 [3.52]	0.79 [3.52]	0.91 [3.87]	0.12 [1.58]	0.17 [2.25]	0.17 [2.20]	0.12 [1.56]	0.14 [1.79]	0.11 [1.38]
	(4)	0.70 [3.02]	0.80 [3.87]	0.85 [3.81]	0.73 [3.51]	0.91 [4.19]	0.21 [2.82]	0.26 [3.45]	0.24 [3.23]	0.20 [2.72]	0.21 [2.75]	0.18 [2.39]
	(5)	0.45 [2.26]	0.62 [3.55]	0.60 [3.03]	0.66 [3.64]	0.78 [4.07]	0.32 [3.74]	0.36 [4.15]	0.38 [4.32]	0.31 [3.57]	0.30 [3.38]	0.26 [2.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCU Quintiles					DCU Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	390	393	392	393	388	37	34	32	33	35	
	(2)	107	107	107	107	107	61	61	60	61	61	
	(3)	76	77	76	76	76	107	109	104	106	108	
	(4)	64	65	64	65	64	231	238	230	232	231	
(5)	59	59	59	59	59	1387	2114	1956	2138	1580		

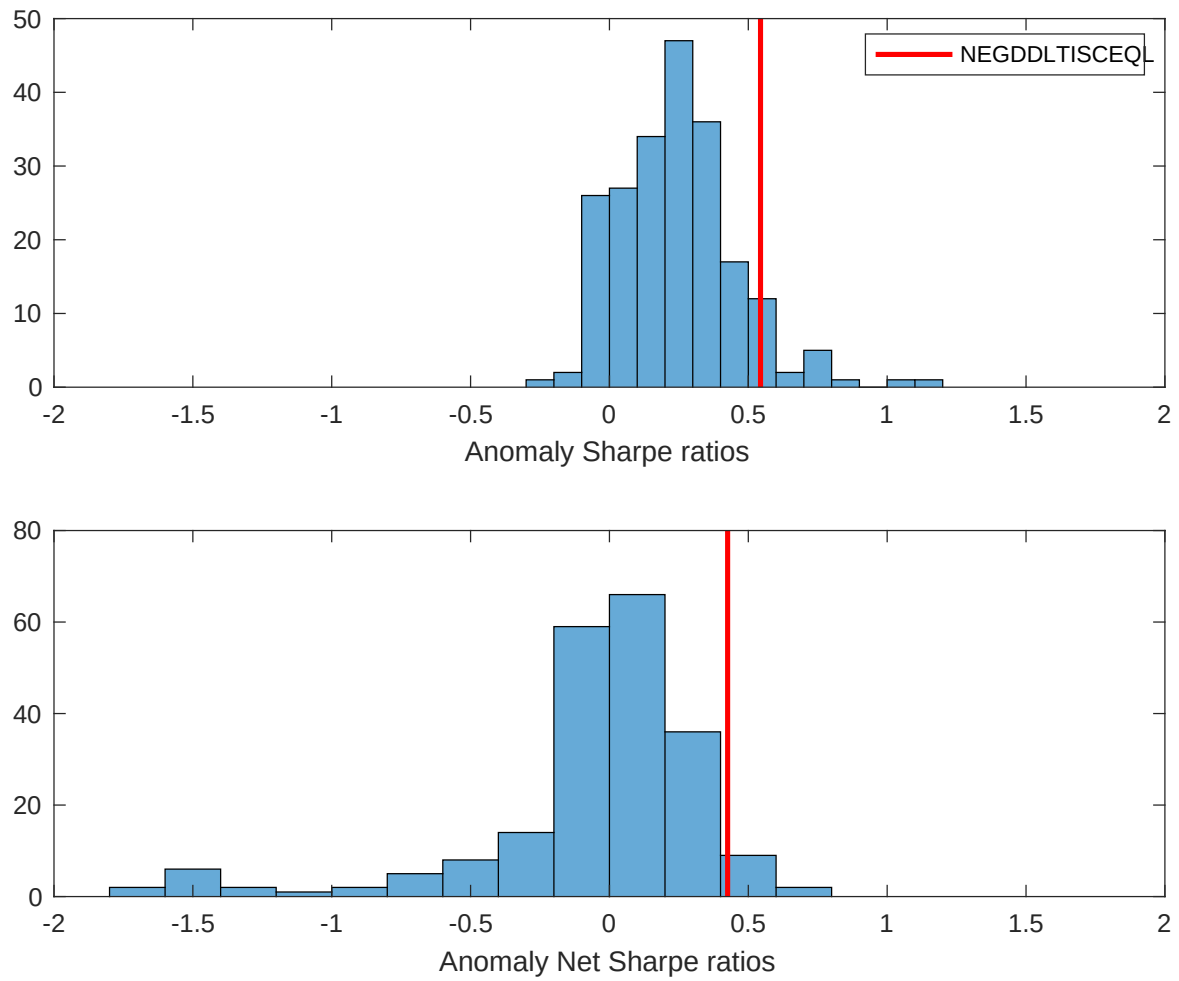


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

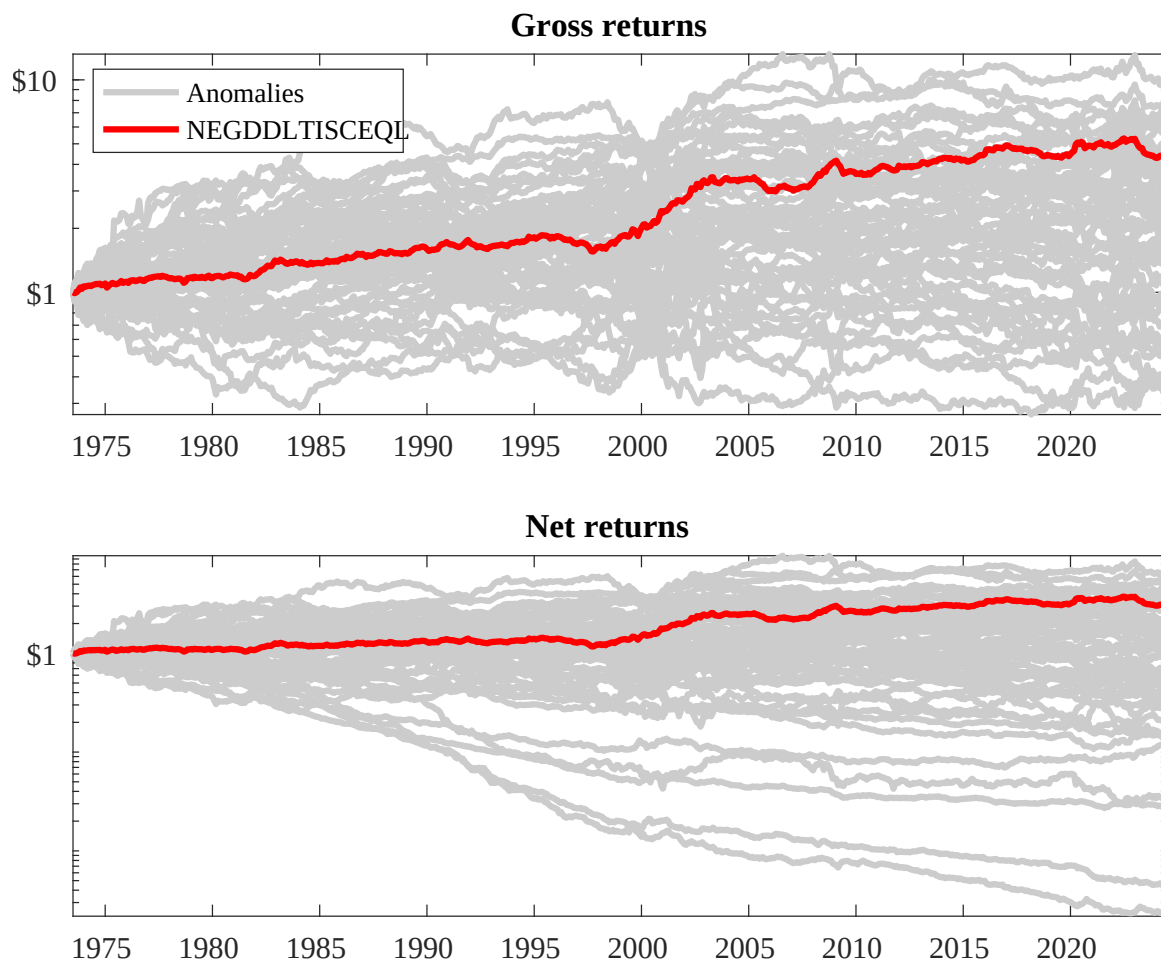
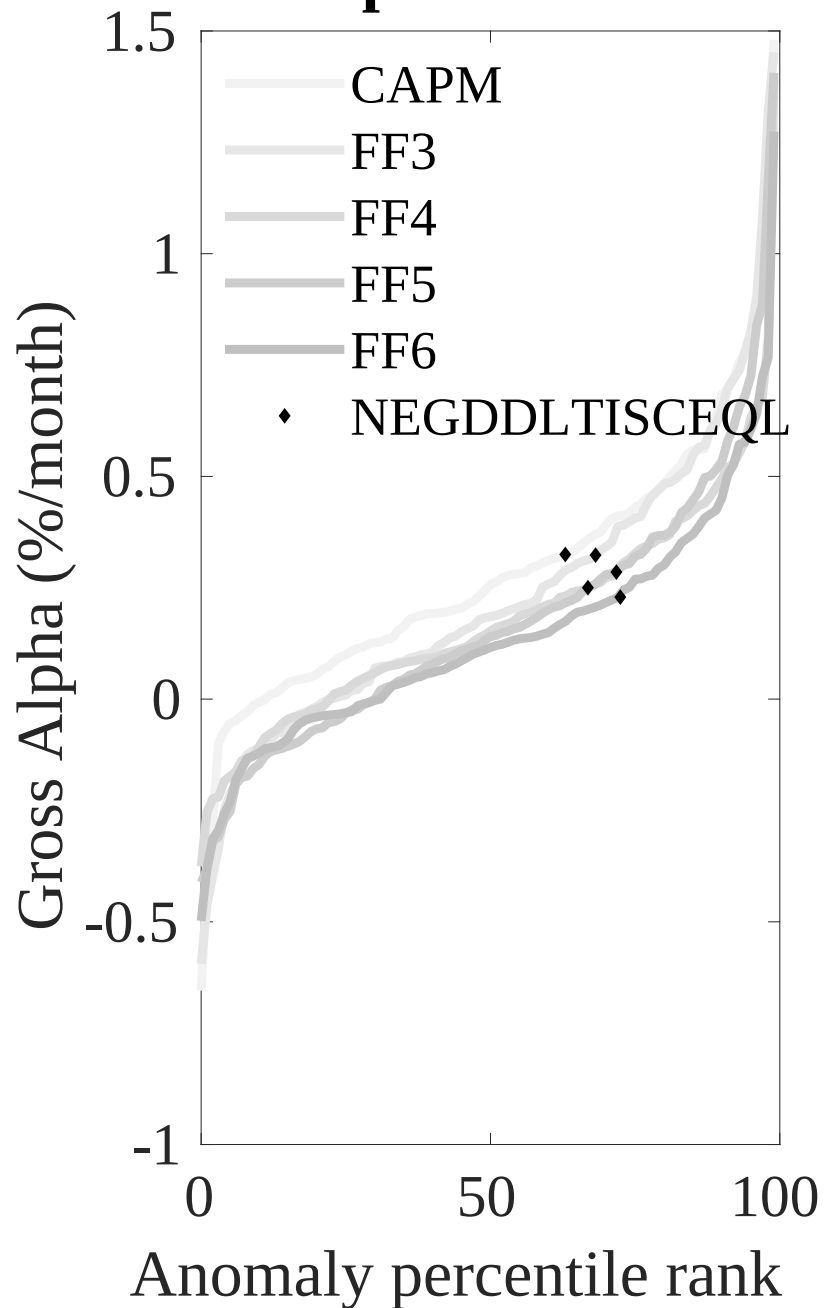


Figure 3: Dollar invested.

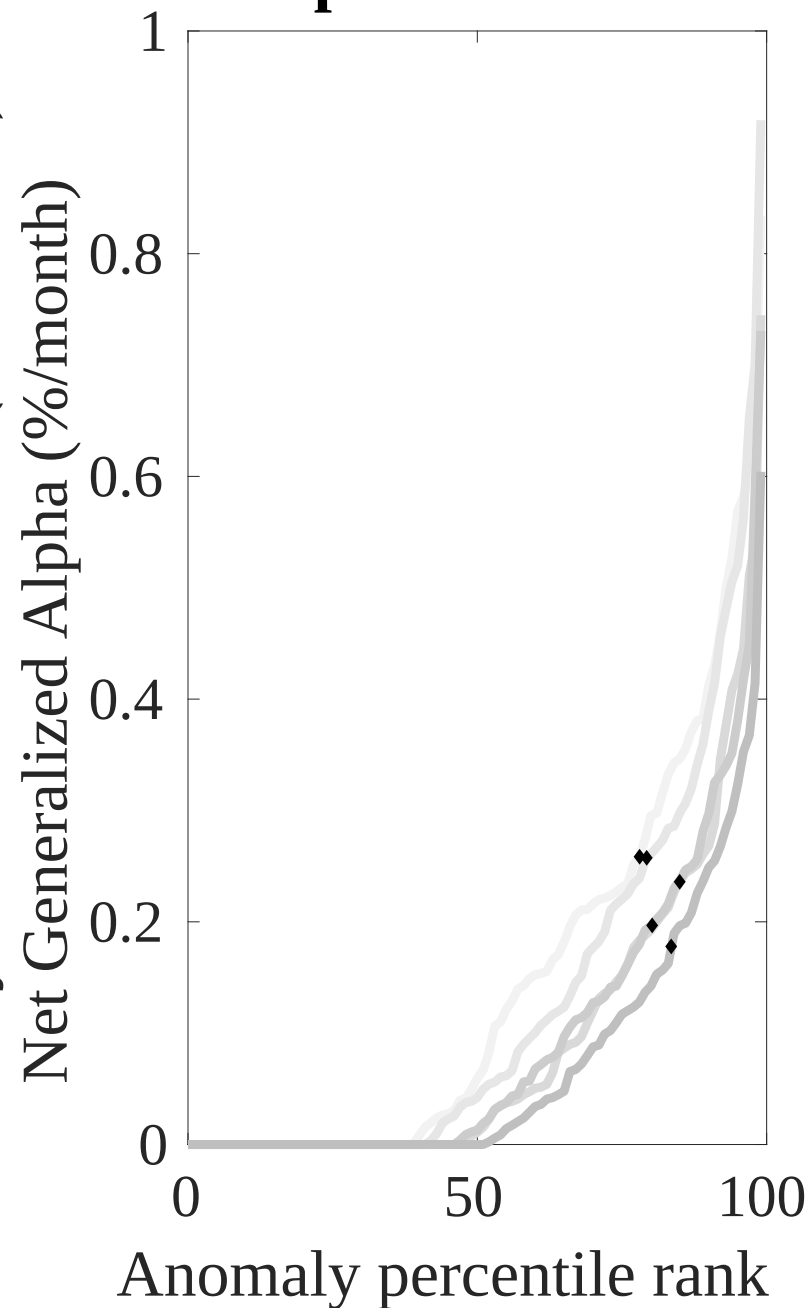
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



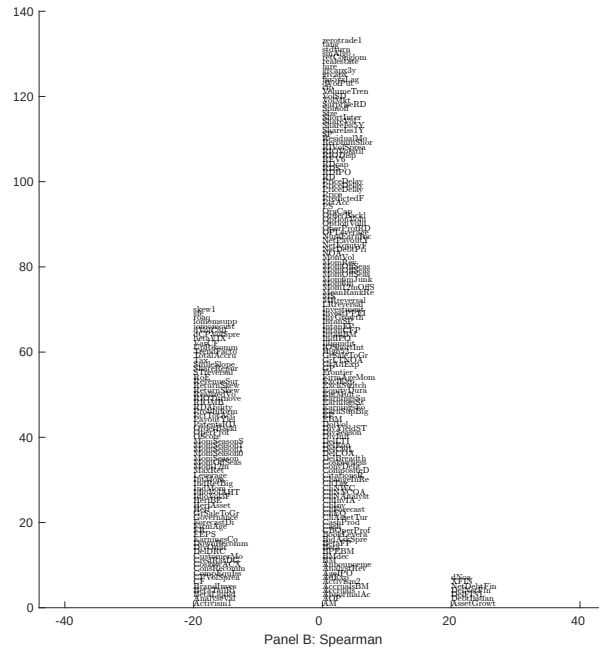
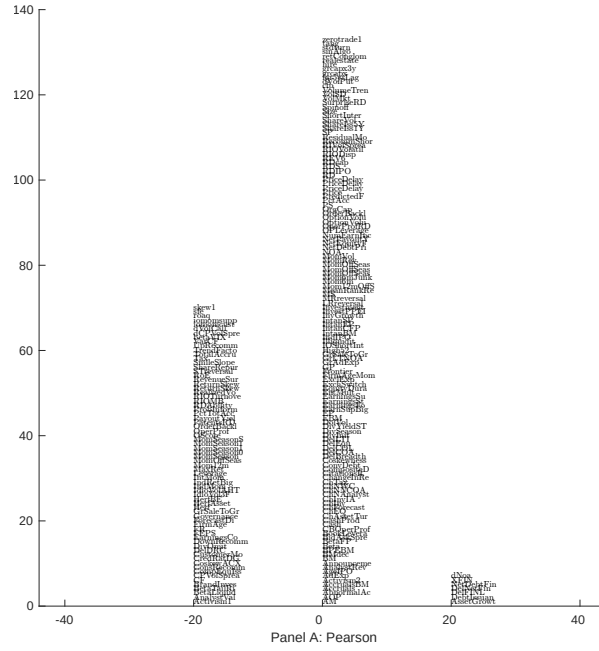


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

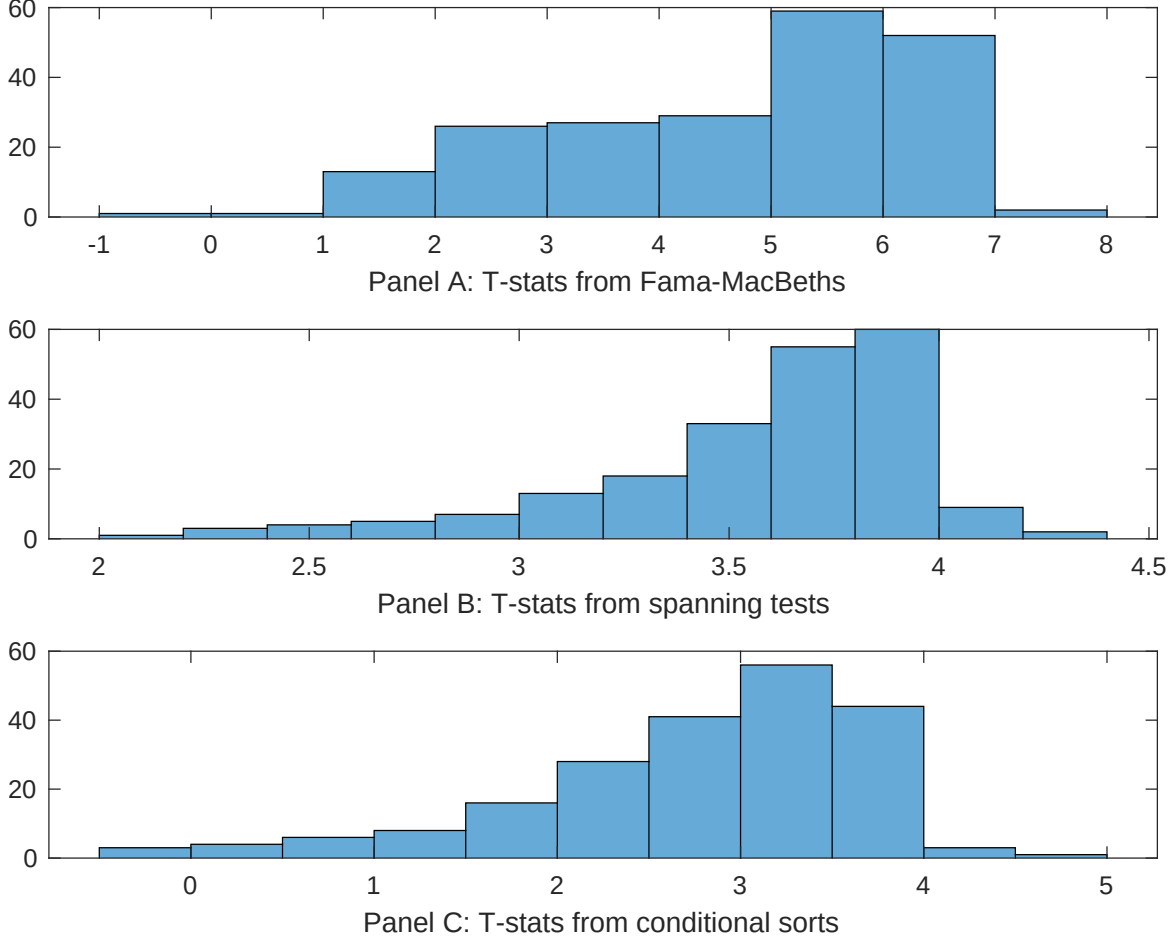


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.69]	0.14 [5.80]	0.14 [5.30]	0.14 [5.66]	0.15 [5.70]
DCU	0.48 [1.98]	0.56 [2.22]	0.73 [2.75]	0.38 [1.52]	0.11 [4.22]	0.34 [1.44]	0.79 [0.27]
Anomaly 1	0.17 [9.27]						-0.11 [-2.30]
Anomaly 2		0.20 [8.82]					0.99 [1.50]
Anomaly 3			0.19 [6.37]				0.98 [1.68]
Anomaly 4				0.11 [8.69]			0.41 [1.80]
Anomaly 5					0.42 [7.14]		0.43 [0.78]
Anomaly 6						0.14 [9.88]	0.95 [5.25]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCU} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the ? five-factor model augmented with the ? momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.20 [2.95]	0.20 [3.01]	0.20 [2.95]	0.21 [3.15]	0.22 [3.25]	0.21 [3.07]	0.20 [2.98]
Anomaly 1	16.33 [4.27]						10.30 [1.96]
Anomaly 2		18.42 [5.02]					9.53 [1.90]
Anomaly 3			13.91 [4.11]				10.20 [2.82]
Anomaly 4				7.42 [1.87]			2.85 [0.68]
Anomaly 5					6.89 [2.60]		6.46 [2.40]
Anomaly 6						3.96 [1.01]	-6.41 [-1.47]
mkt	-4.72 [-3.04]	-5.06 [-3.28]	-3.19 [-1.96]	-5.10 [-3.25]	-5.22 [-3.33]	-5.08 [-3.23]	-3.57 [-2.21]
smb	-0.28 [-0.12]	-0.23 [-0.10]	5.86 [2.26]	0.86 [0.36]	2.15 [0.90]	1.47 [0.62]	3.16 [1.16]
hml	-10.56 [-3.52]	-10.90 [-3.66]	-10.10 [-3.35]	-11.98 [-3.95]	-11.86 [-3.92]	-11.97 [-3.92]	-9.26 [-3.06]
rmw	-0.17 [-0.06]	-0.73 [-0.24]	-7.01 [-1.92]	1.33 [0.43]	2.30 [0.74]	1.36 [0.44]	-5.93 [-1.61]
cma	23.30 [4.94]	23.99 [5.19]	19.70 [3.90]	19.79 [2.98]	22.64 [4.39]	25.83 [4.76]	11.76 [1.74]
umd	2.10 [1.33]	2.32 [1.50]	3.41 [2.21]	3.67 [2.35]	2.81 [1.77]	3.38 [2.15]	1.53 [0.97]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	14	15	14	12	13	12	17

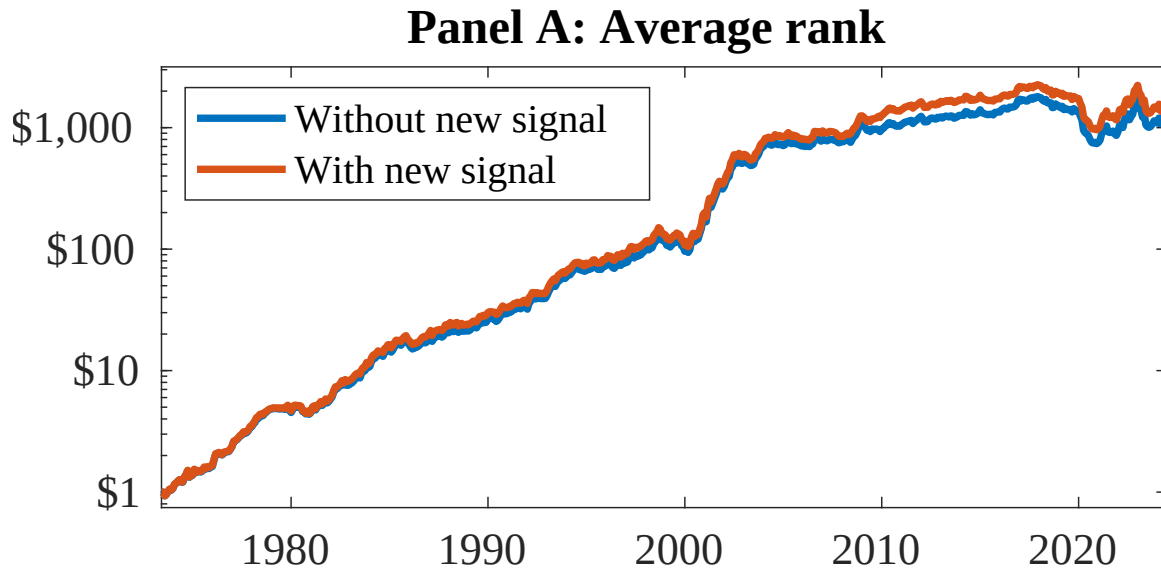


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following ?. In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCU. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.